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IOT (Internet Of Things)

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CERTIFICATE

This is to certify that Mr./Ms. Raj Dilip Yenare of S.Y. Computer Science (SYCS) Roll No. __2524509 has satisfactorily completed The Internet Of Thing Mini Project smart quality enspection_ during the academic year 2025 – 2026, as a part of the practical requirement. The project work is found to be satisfactory and is approved for submission.

PROF. INCHARGE **HEAD OF DEPT**

Sahil shukla_ **Laxmi jeswani__**

SR.NO	CHAPTER	PG.

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1. Introduction

Ensuring food quality and safety is a major challenge in homes, restaurants, and food processing industries. Traditional quality inspection methods often rely on manual observation, which can be time-consuming, inconsistent, and prone to human error. With the advancement of Internet of

Things (IoT) technology and embedded systems, automated and low-cost solutions are becoming increasingly accessible.

This project focuses on developing a **Smart Food Quality Inspection System using ESP32-CAM**. The ESP32-CAM is a compact, low-cost microcontroller with an integrated camera and Wi-Fi capability. It enables real-time image capture and wireless data transmission, making it ideal for smart monitoring applications.

The system captures images of food items and analyzes them to detect visible quality indicators such as:

- Color changes (e.g., browning, mold growth)
- Surface texture abnormalities
- Shape deformation
- Spoilage signs

By applying basic image processing or machine learning techniques, the ESP32-CAM can classify food as **fresh or spoiled** based on predefined parameters. The captured data can be transmitted to a web server or mobile application for monitoring and record keeping.

This smart inspection system offers several advantages:

- Low cost and compact design
- Real-time monitoring
- Reduced human intervention
- Improved consistency in quality checks

Overall, the Smart Quality Inspection System using ESP32-CAM demonstrates how IoT and embedded vision technology can enhance food safety, reduce waste, and improve efficiency in quality control processes.

2. Requirement Specification

1. Hardware Requirements

ESP32 camera module

Micro USB cable

Esp 32 cam connector

Internet connectivity (Wi-Fi network)

2. Software Requirements

Arduino IDE (Version 1.8.19 recommended)

ESP32 Board Package installed in Arduino IDE

Required Libraries (WiFi, ThingSpeak Library)

ThingSpeak Cloud Account – **ThingSpeak**

Operating System: Windows/Linux/macOS

3. Functional Requirements

The functional requirements describe what the system must do to perform food quality inspection effectively.

1.image capture

- The system shall capture images of food items using the ESP32-CAM module.
- The system shall allow automatic image capture at predefined time intervals.
- The system shall allow manual image capture through a user command (button or web interface).

2.image processing

- The system shall process captured images to analyze visible quality indicators such as:
  - Color changes
  - Texture variations
  - Presence of mold or dark spots
- The system shall convert images into suitable formats for analysis (e.g., grayscale or resized images).
- The system shall compare captured images with predefined quality parameters or trained model data.

3.quality classification

- The system shall classify food items into categories such as:
  - Fresh
  - Acceptable
  - Spoiled
- The system shall generate a decision result based on analysis results.

4.data transmission

- The system shall transmit inspection results via Wi-Fi.
- The system shall display results on a web page or mobile interface.
- The system shall store inspection results for future reference (optional cloud or local storage).

5.user interface

- The system shall provide a web-based dashboard for:
  - Viewing live camera feed

    - Viewing captured images
    - Checking inspection results
- The system shall allow users to configure inspection intervals and threshold values.

6.alert and notification

- The system shall send alerts when spoiled food is detected.
- The system shall display warning messages on the dashboard.
- The system may send notifications via email or mobile app (optional).

7.system controll

- The system shall connect automatically to a configured Wi-Fi network.
- The system shall allow resetting or restarting remotely.
- The system shall indicate system status using LED indicators.

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4. Non-Functional Requirements

The system should provide accurate and stable readings.

Data transmission should be reliable and secure.

The system should operate continuously with minimal maintenance.

The solution should be low-cost and energy-efficient.

3. System Design

- **ESP32-CAM**
- **ESP32-CAM USB Programmer / ESP32-CAM-MB (Connector Board)**
- **USB Data Cable**
- **PC**

You can still make a **basic Smart Food Quality Inspection System** using only the camera and image processing.

📌 Project Topic
"Smart Food Quality Inspection Using ESP32-CAM"

(Basic Version – Camera-Based Detection)

📊 Block Diagram (For Your Available Components)

```
Power from PC (USB)
        ↓
ESP32-CAM
        ↓
Image Capture (Food Image)
        ↓
Image Processing (Color/Texture Analysis)
        ↓
Quality Decision (Fresh / Spoiled)
        ↓
Web Page Display (Result)
```

◈ Block Explanation

1 Power Supply

USB cable connected to PC supplies 5V power.

2 ESP32-CAM

Main controller with built-in Wi-Fi and camera.

3 Image Capture

Camera captures image of food (fruit, vegetable, etc.).

4 Image Processing

Using:

- Arduino IDE (basic color detection), OR
- Web server streaming and manual inspection

You can detect:

- Color change (brown spots, dark patches)
- Surface texture differences

5 Quality Decision

Based on color values (RGB):

- Bright/Normal color → Fresh
- Dark/Brown/Black → Spoiled

6 Web Output

ESP32 hosts a webpage:

- Shows live camera feed
- Displays result (Fresh / Spoiled)

💡 Simple Project Idea (Very Easy to Implement)

Instead of AI, do:

✓ Capture image
✓ Convert to RGB values
✓ Compare brightness
✓ Print result on web page

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4. Implementation

CODE :

```
#include "esp_camera.h"


// Camera model (AI Thinker)
#define CAMERA_MODEL_AI_THINKER


#include "camera_pins.h"


void setup() {
  Serial.begin(115200);
  Serial.println("Food Quality Inspection Started");


  camera_config_t config;
  config.ledc_channel = LEDC_CHANNEL_0;
  config.ledc_timer = LEDC_TIMER_0;
  config.pin_d0 = Y2_GPIO_NUM;
  config.pin_d1 = Y3_GPIO_NUM;
```

```
  config.pin_d2 = Y4_GPIO_NUM;
  config.pin_d3 = Y5_GPIO_NUM;
  config.pin_d4 = Y6_GPIO_NUM;
  config.pin_d5 = Y7_GPIO_NUM;
  config.pin_d6 = Y8_GPIO_NUM;
  config.pin_d7 = Y9_GPIO_NUM;
  config.pin_xclk = XCLK_GPIO_NUM;
  config.pin_pclk = PCLK_GPIO_NUM;
  config.pin_vsync = VSYNC_GPIO_NUM;
  config.pin_href = HREF_GPIO_NUM;
  config.pin_sscb_sda = SIOD_GPIO_NUM;
  config.pin_sscb_scl = SIOC_GPIO_NUM;
  config.pin_pwdn = PWDN_GPIO_NUM;
  config.pin_reset = RESET_GPIO_NUM;
  config.xclk_freq_hz = 20000000;
  config.pixel_format = PIXFORMAT_GRAYSCALE;

  config.frame_size = FRAMESIZE_QQVGA;
  config.jpeg_quality = 12;
  config.fb_count = 1;

  if (esp_camera_init(&config) != ESP_OK) {
    Serial.println("Camera Init Failed");
    return;
  }
}

void loop() {

  camera_fb_t * fb = esp_camera_fb_get();
```

```
  if (!fb) {
    Serial.println("Capture Failed");
    return;
  }

  long totalBrightness = 0;

  for (int i = 0; i < fb->len; i++) {
    totalBrightness += fb->buf[i];
  }

  int avgBrightness = totalBrightness / fb->len;

  Serial.print("Average Brightness: ");
  Serial.println(avgBrightness);

  if (avgBrightness < 60) {
    Serial.println("Food Quality: SPOILED");
  } else {
    Serial.println("Food Quality: FRESH");
  }

  Serial.println("-----------------------");

  esp_camera_fb_return(fb);

  delay(5000);  // Check every 5 seconds
}
```

5.3 Testing Image

6. Future Scope and Conclusion

6.1 Future Scope

The current system uses basic brightness analysis to detect food freshness. In future, the system can be improved in the following ways:

1 AI-Based Detection

Instead of simple brightness comparison, a **Machine Learning model** can be integrated to:

- Detect fungus, mold, black spots
- Identify ripeness level
- Classify multiple food types (fruits, vegetables, meat)

This will increase accuracy and reliability.

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2 Cloud Integration (IoT Based System)

The system can be connected to:

- Cloud storage
- Mobile application
- Web dashboard

This allows:

- Remote monitoring
- Data storage
- Real-time alerts

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3 Multi-Sensor Integration

Additional sensors can be added:

- Temperature sensor
- Humidity sensor
- Gas sensor (for detecting spoilage gases)

This will make the system more intelligent and suitable for warehouses.

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4 Industrial Automation

The project can be expanded for:

- Conveyor belt food inspection
- Automatic rejection of spoiled items
- Smart packaging units

This will be useful in food processing industries.

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5 Mobile Application Development

A dedicated mobile app can:

- Show live camera feed
- Send notification alerts
- Store inspection history

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6 Higher Resolution and Deep Learning

Using advanced boards like:

- Raspberry Pi
- NVIDIA Jetson Nano

Can support:

- Deep learning models
- Object detection
- Defect localization

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7 Smart Refrigerator Integration

The system can be integrated into:

- Smart refrigerators
- Retail stores
- Supermarkets

To automatically monitor food freshness.

6.2 Conclusion

The Smart Food Quality Inspection System using ESP32-CAM successfully demonstrates a low-cost and simple method to monitor food freshness. The system captures images of food items and analyzes brightness values to classify them as fresh or spoiled. It provides a basic real-time inspection solution without requiring additional hardware components. This project proves that ESP32-CAM can be effectively used for image-based quality monitoring and can be further enhanced with AI and IoT technologies for higher accuracy and industrial applications.

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7. References

The Smart Food Quality Inspection System using ESP32-CAM successfully demonstrates a low-cost and simple method to monitor food freshness. The system captures images of food items and analyzes brightness values to classify them as fresh or spoiled. It provides a basic real-time inspection solution without requiring additional hardware components. This project proves that ESP32-CAM can be effectively used for image-based quality monitoring and can be further enhanced with AI and IoT technologies for higher accuracy and industrial applications.

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8. Glossary

ESP32-CAM
A low-cost microcontroller module with a built-in Wi-Fi and camera used for image capture and processing.

Microcontroller
A small computer on a single chip that controls electronic devices.

IoT (Internet of Things)
A system where devices are connected to the internet to collect and exchange data.

Image Processing
A technique used to analyze and manipulate images to extract useful information.

Grayscale Image
An image that contains only shades of gray (black to white) instead of colors.

Brightness
The intensity of light in an image, used to determine food freshness in this project.

Threshold Value
A predefined value used to classify food as fresh or spoiled.

Pixel
The smallest unit of a digital image.

Wi-Fi Module
A wireless communication feature in ESP32 used to send and receive data.

Serial Monitor
A tool in Arduino IDE used to display output data from the microcontroller.

QQVGA Resolution
A low image resolution (160×120 pixels) used to reduce memory usage.

Firmware
The software programmed into the microcontroller to control its operation.